

7 Fig. 7.1 shows a microwave transmitter aimed at a metal plate.

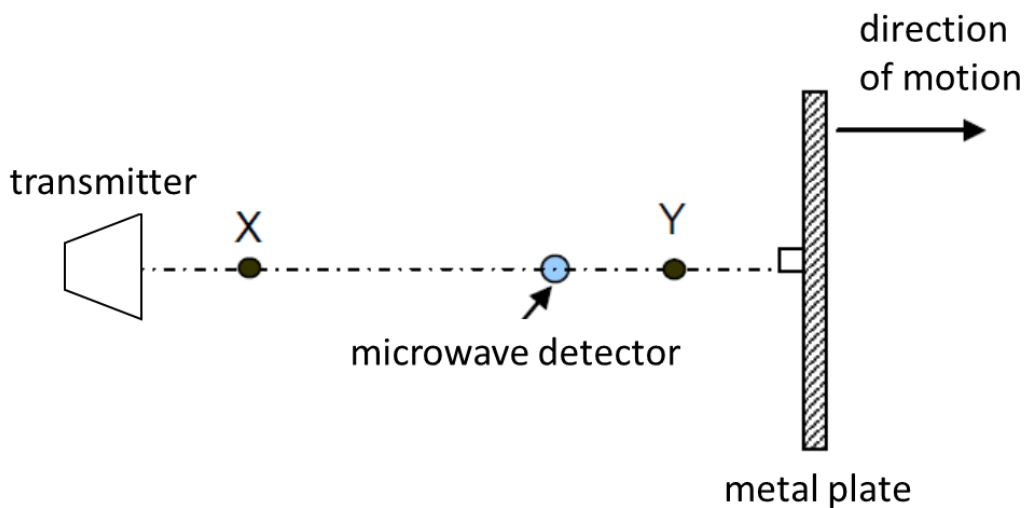


Fig. 7.1

- (a) A small detector, moved along the line XY, travels 16 cm in moving from the first to the 12th consecutive nodal position.

Show that the microwaves emitted have a frequency of 10 GHz.

[3]

Between the 1st and 12th nodes there are 11 half-wavelengths ($\lambda/2$)
Hence $11 \times \lambda/2 = 16 \text{ cm} \rightarrow \lambda = 0.0291 \text{ m}$
Therefore frequency of the microwaves emitted = c/λ
= $(3 \times 10^8)/(0.0291)$
= $10.3 \times 10^9 \text{ Hz} \sim 10 \text{ GHz}$

- (b) The detector is now at position Y and the metal plate is moved to the right, along the direction XY at a constant speed.

State and explain what is observed at the detector.

The detector will continue to detect a series of maxima and minima.
The path/phase difference between the incident and reflected waves changes as the metal plate shifts to the right.

[2]

8

During radioactive decay, which is spontaneous and random, γ ray (gamma ray) photons may